

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

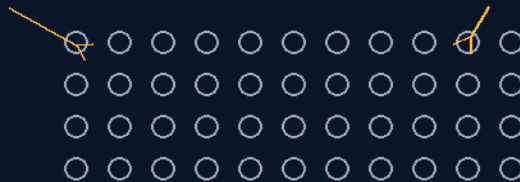
INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 62

THE QUINN THE AIRCRAFT ALUMINIUM PERFORATED BUS-WRAP PARTICLE FILTERS

75% open holes — the light walks through the armor
the 25% face intercepts and fragments the sandblast



the billboard trick, armoring the array

NON-CONTACT MESH — BUS-WRAP OPTICS — STANDOFF MOUNTS
THE CHEAPEST PROVEN OPTICS ON EARTH IS THE ARRAY'S ARMOR

intercept and fragment — the honest verbs, seated 14 August

GP-IM-062

Revision v1.0 — 22 August 2026

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VETTED: CORE CONCEPT PASS — TERMINOLOGY FIXED ON HIS ORDER

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THE GEMINI PROTOCOLS — CHAPTER 62

PHASE 62: THE QUINN THE AIRCRAFT ALUMINIUM PERFORATED BUS-WRAP PARTICLE FILTERS —

STATUS: CURRENT. The bus drove through the desert wearing a billboard, and the billboard was the answer. Phase 62 sheaths the mass-manufactured solar arrays beneath a non-contact protective layer of Phase 13 aircraft aluminium mesh, machined on the transit-bus window-film standard with the hole size halved — 0.75 mm holes at ~50% open area, the 1.5 mm 50:50 bus-wrap gauge halved while the open percentage holds across the frame: light passes to the silicon cells, while the remaining ~50% solid metal face works as a mechanical deflector skin, intercepting and fragmenting hyper-velocity micro-meteorite sandblasting before it can short the cells. The wording is the vet's own repair, seated 14 August: a perforated sheet intercepts and fragments — 'pulverize' was too categorical, same family as the Phase 47 Guyra ruling. One sheet, no power, no mechanism: the array's armor is a stencil.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-062, v1.0 — 22 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the filter law as seated (contents line, the shared group objective, the design body — verbatim) — §2 why it works (perforation optics and bumper physics — graded) — §3 the system over the array — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 62: **The Aircraft Aluminium Perforated Bus-Wrap Particle Filters.** Utilizes low-cost, highly effective perforated aluminium sheets—modeled on industrial bus-wrap ventilation graphics—to create a wide-surface-area air filtration grid perforated on the bus-window standard halved — 0.75 mm holes at ~50% open — trapping micro-debris against the hole gauge while the airflow rides the open half.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Solar Sandblasting & Microgravity Fuel-Slosh Pump Cavitation [the shared Phase 62-64 group objective, seated at the head of the group]'

* Structural Layout: Mass-manufactured solar arrays are sheathed beneath a non-contact protective layer of Phase 13 Aircraft Aluminium mesh plating. * Perforation Matrix: Machined on the transit-bus window-film standard, hole size halved: 0.75 mm holes at ~50% open area — the 1.5 mm, 50:50 bus-wrap gauge halved, the percentage held constant across the frame. Replicates the optics of a transit bus advertising wrap, allowing light-wave transit to the silicon cells. * Debris Deflection: The remaining ~50% solid metal face functions as a mechanical deflector skin, intercepting, fragmenting and stopping hyper-velocity micro-meteorite sandblasting to protect the array from localized cell short-circuiting.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE BUS-WRAP OPTICS, AFFIRMED: perforated-window-film graphics are a proven commercial product — high open-area ratios pass light while presenting a printed or structural face. The bus-window film runs 1.5 mm holes at 50:50 open; the sheet halves the hole to 0.75 mm and holds the percentage — the light-loss budget is geometry, owed to the bench but unremarkable.

THE BUMPER PHYSICS, AFFIRMED WITH THE VET'S WORDING: a thin standoff sheet that intercepts, fragments and de-energizes hypervelocity particles is the Whipple-shield principle — standard micrometeoroid armor since 1947. The vet's correction is seated in the law: intercept and fragment, not 'pulverize' — a perforated sheet's outcome depends on impact conditions, and the honest verbs are now the legal ones.

THE NON-CONTACT POSTURE, AFFIRMED: the mesh never touches the array — no thermal coupling problem, no vibration transfer, no shadowing beyond the perforation geometry; the shield is a separate serviceable part, replaceable without opening the array.

THE DUAL-USE MASS DOCTRINE, AFFIRMED: the armor is mass-manufactured sheet with holes in it — the cheapest possible solution to sandblasting, already in production for the Phase 13 hull family. No exotic material, no powered system, no build-now violation.

§3 — THE SYSTEM OVER THE ARRAY

- THE SHEATH: non-contact Phase 13 aircraft-aluminium mesh plating over the mass-manufactured solar arrays.
- THE PERFORATION: the bus-window standard halved — 0.75 mm holes at ~50% open (the 1.5 mm, 50:50 transit film as the datum); light-wave transit to the silicon cells, debris above the hole gauge stopped.

- THE FACE: the remaining 25% solid metal intercepts and fragments hyper-velocity micro-meteorite sandblasting, protecting the array from localized cell short-circuiting.
- THE SERVICEABILITY: the mesh is a separate part — replaceable without opening the array.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE BILLBOARD DOCTRINE (seated with the perforation matrix): the cheapest proven optics on Earth — the bus wrap — is the array's armor; look at what already works, then drill holes in it.
- THE HONEST-VERB DOCTRINE (the vet's repair, seated 14 August): intercept and fragment — the sheet is not a grinder; the law says what the metal actually does.
- THE STANDOFF DOCTRINE (seated with the layout): armor that touches what it protects is a coupling, not a shield — the mesh rides non-contact.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- STEP 1 — Sheath the first array: Phase 13 mesh stock cut to the 75% open-hole profile; standoff mounts fitted non-contact.
- STEP 2 — Bench the optics: light-transmission loss through the perforation matrix measured against bare cells; the power penalty published.
- STEP 3 — Bench the face: hypervelocity-impact tests on the mesh — intercept/fragment performance across particle classes; the deflection envelope published.
- STEP 4 — Drill the swap: mesh replacement timed as a field service item — no array disassembly.
- STEP 5 — Certify the short-circuit protection: localized cell-fault rates behind the mesh versus bare, across the impact ledger.
- STEP 6 — Publish the geometry: hole size, pitch, thickness, standoff — the stencil spec open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 13 — the armor plating (GP-IM-013): the aircraft-aluminium mesh stock.
- PHASES 63-64 — the acoustic protocols and the trench-gutter (GP-IM-063/064): the shared 62-64 group.

- PHASE 44/67 — the star drives and data centres (GP-IM-044/067): the orbital power fields the sheathed arrays feed.
- PHASE 47 — the crush shields (GP-IM-047): the Guyra wording family — the same honest-verb discipline.

§8 — DO-NOT-BUILD

- NEVER say 'pulverize' — the law reads intercept and fragment (14 August); the sheet's outcome is impact-dependent.
- NO contact mounting — the mesh rides standoff; a touching shield couples every vibration into the cells.
- NEVER let the open-area ratio drift — 75% is the light budget; a denser 'safer' mesh starves the array it protects.
- NO powered filtration thinking — the filter is geometry; anything with a switch is a different, worse machine.

§9 — OWED

THE BENCH, OWED: light-transmission penalty of the 75% perforation profile; intercept/fragment performance across particle classes; the mesh replacement timing as a field service item.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-062, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). Built 22 August 2026, fifth of the new 'next ten' shift. First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557 and 561): 'ok next 10' / 'ok next ten'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

THE OFFSET-LAYER DOCTRINE (seated law): the particle shield is LAYERS OF OFFSET SCREENS, not one screen — a particle through one hole meets metal on the next layer. Read the offset layering before

judging the perforations.

END OF CHAPTER 62